

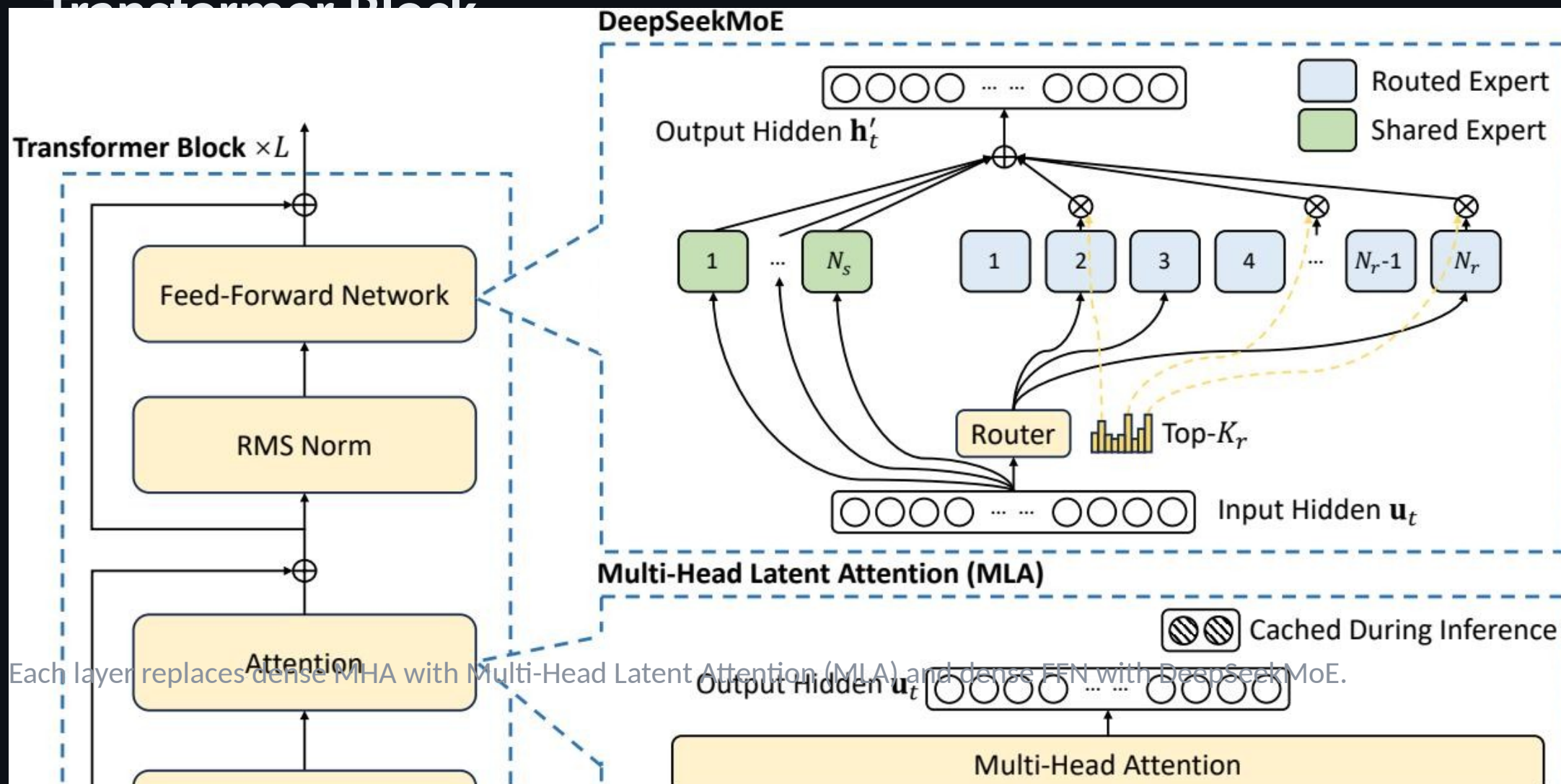
DeepSeek-V2: A Strong, Economical, and Efficient MoE Language Model

236B total parameters · 21B activated per token · 128K context length

■ The LLM Scaling Dilemma: Performance vs. Training Cost and Inference Speed

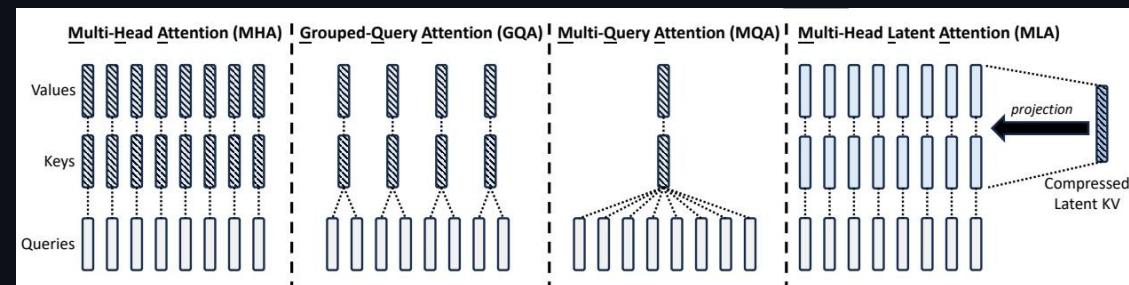
- Larger dense models improve capability but scale costs super-linearly in training FLOPs and memory.
- Autoregressive inference is bottlenecked by the KV cache: full key-value vectors per head per token explode memory bandwidth.
- Existing shortcuts (MQA, GQA) shrink cache but measurably degrade attention capacity.
- MoE promises sparse activation, yet communication overhead and load imbalance often erode the savings.
- Need: an architecture that is simultaneously cheaper to train, faster to generate, and stronger at evaluation.

DeepSeek-V2 Architecture: MLA + DeepSeekMoE in a Transformer Block



Multi-Head Latent Attention Compresses KV Cache by 93.3%

- Instead of caching full K and V vectors per head, MLA compresses them into a compact latent vector c_t^{KV} .
- Full keys, values, and a decoupled RoPE-based positional key are reconstructed on-the-fly via learned projections.
- Result: better performance than standard MHA with dramatically less cache — strictly dominating GQA/MQA.
- KV cache reduction: 93.3% vs. conventional MHA.



■ DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

- Shared experts (N_s): always activated, capture common knowledge across all tokens.
- Routed experts (N_r): sparsely activated via a top- K_r router, enabling massive parameter counts with modest per-token FLOPs.
- Fine-grained expert segmentation increases specialization potential without increasing active parameters.
- Device-limited routing constrains each token to experts on at most M devices, cutting cross-node communication.
- Auxiliary losses enforce load balance across experts and devices; token dropping during training prevents overflow without affecting inference.

■ 42.5% Training Cost Reduction and 5.76× Generation Throughput vs. DeepSeek 67B

DeepSeek 67B (Dense)

Baseline dense predecessor

DeepSeek-V2 (MoE + MLA)

Training cost: ↓ 42.5%

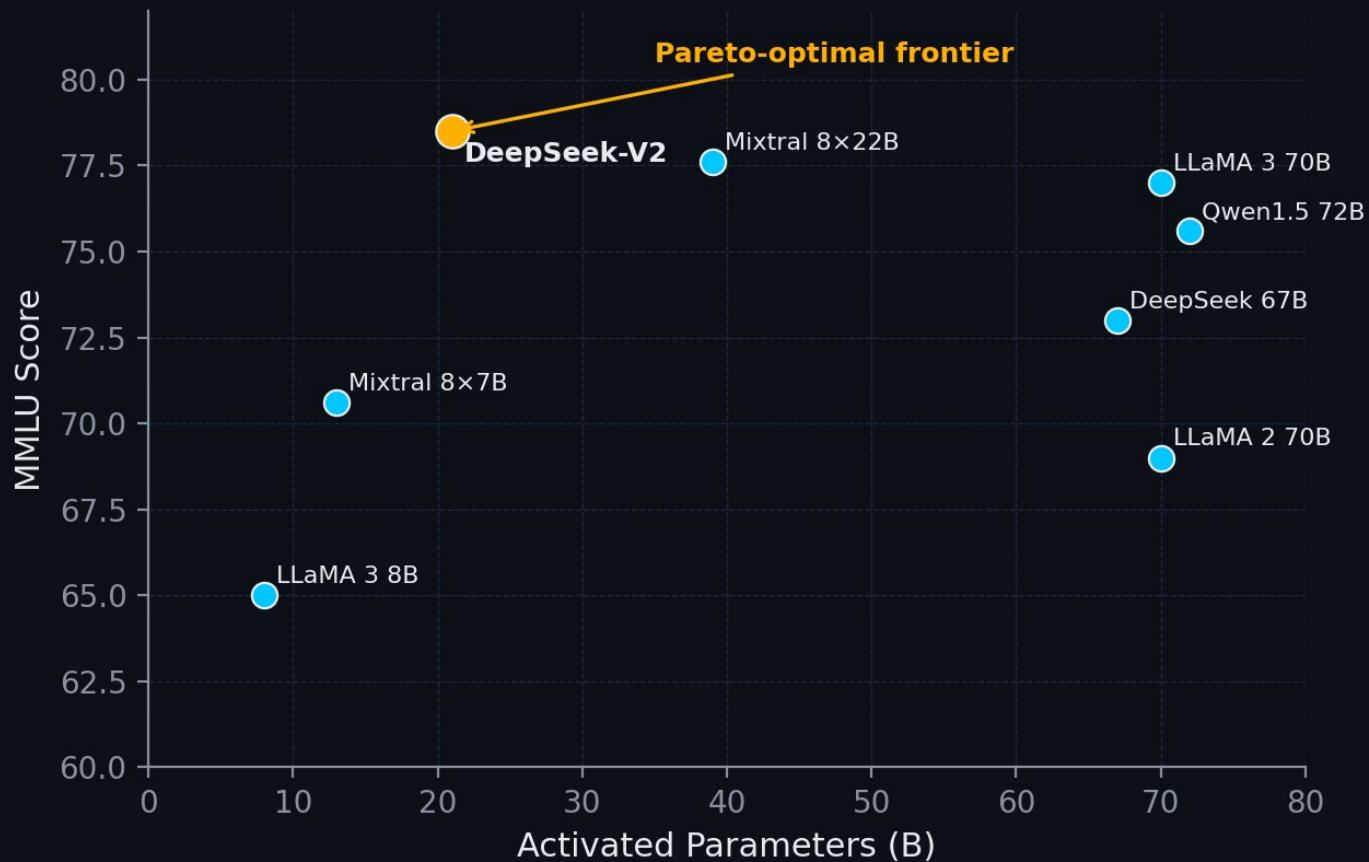
KV cache size: ↓ 93.3%

Max generation throughput: ↑ 5.76×

All metrics are relative improvements over the DeepSeek 67B dense baseline. No absolute FLOPs or cache sizes are stated.

DeepSeek-V2 Tops MMLU Among Open-Source Models at Only 21B

Activated Parameters



Model	Act. Params	MMLU
DeepSeek-V2	21	78.5
LLaMA 3 70B	70	77.0
Mixtral 8×22B	39	77.6
Qwen1.5 72B	72	75.6
DeepSeek 67B	67	73.0
LLaMA 2 70B	70	69.0
Mixtral 8×7B	13	70.6
LLaMA 3 8B	8	65.0

Lower-right = more efficient. DeepSeek-V2 is Pareto-optimal.

■ DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchmarks

- Alignment pipeline: Supervised Fine-Tuning on 1.5M conversational sessions, followed by GRPO-based Reinforcement Learning.
- AlpacaEval 2.0 (length-controlled win rate): 38.9%
- MT-Bench (overall): 8.97
- AlignBench (overall, Chinese): 7.91 — outperforming all open-source and most closed-source models in Chinese evaluation.
- Demonstrates that architectural efficiency and post-training quality are not mutually exclusive.

■ Key Takeaways: Architectural Innovation Unlocks Efficient Scaling

1. MLA is a strictly better KV-cache solution: 93.3% reduction with superior performance to MHA, dominating GQA/MQA.
2. DeepSeekMoE enables economical training of a 236B-parameter model at 42.5% lower cost than a 67B dense model.
3. Only 21B activated parameters match or exceed 70B-class dense models on MMLU — a Pareto-optimal trade-off.
4. 5.76× throughput improvement over DeepSeek 67B makes large-scale deployment practical.
5. DeepSeek-V2 Chat (RL) is the strongest open-source chat model in Chinese and competitive with closed-source systems in English.